# Fangwei Chang

University of Toronto
fangwei.chang@mail.utoronto.ca

Google Scholar: link to publications
linkedin.com/in/fangwei-c


## Education

**University of Toronto** — ***Toronto, Canada***
***PhD Candidate**, Electrical and Computer Engineering (ECE)* — ***Sep 2021 — Present***
- **Thesis:** Near Field Wireless Power Transfer and Focusing using Electromagnetic Metamaterials
- Using numeric methods, optimization, and finite-element simulation tools to design and optimize relevant antenna structures, e.g. COMSOL, HFSS, and the method of moments (MoM). Designs are fabricated in-lab using laser printed circuit board (PCB) prototyping machine. Experimental verification is performed using relevant RF measurement setups.
- **Available for full-time positions starting Nov 2026**.

**University of Toronto** — ***Toronto, Canada***
***Master of Applied Science (MASc)**, Electrical and Computer Engineering (ECE)* — ***Sep 2019 — Sep 2021***
- **Thesis:** Shifted-beam Array of Coils for Highly Focal Transcranial Magnetic Stimulation, *link*

**University of Toronto** — ***Toronto, Canada***
***Honors Bachelors of Science (BSc Hons)**, Physics & Mathematics Double Major* — ***Sep 2015 — Jun 2019***
- Dean's List Award recipient 2016-2018.

## Technical Skills

| | |
|---|---|
| **Programming Languages:** | Python, MATLAB, C++, Java |
| **Simulation Tools:** | COMSOL Multiphysics, Ansys Electronics Desktop (AEDT: HFSS, Icepak), Keysight Advanced Design System (ADS) |
| **Modeling/Numerical Methods:** | Finite-element method (FEM), finite-difference time-domain (FDTD), method of moments (MoM), reduced order modeling/surrogate modeling, multiphysics modeling, optimization |
| **AI/ML Tools:** | Tensorflow, Keras, PyTorch |
| **Software Tools:** | Anaconda, Jupyter Notebook, PyCharm, VSCode, LaTeX, Git, high performance computing (HPC) |
| **Subject Areas:** | Electromagnetics, metamaterials, physics, wireless power transfer |
| **Hardware:** | Radiofrequency (RF) design, RF equipment, antenna design/measurements, printed circuit board (PCB) prototyping, laser etching, 3-D printing, composite materials, machining |

## Professional Experience

**Research & Development Intern** — ***Remote, Canada***
***Synopsys**: Electronics, Simulation, and Optics Unit* — ***May 2025 - Apr 2026***
- **Developed, implemented, and tested** methods for enhancing the prediction accuracy of surrogate thermal models for integrated circuit (IC) packages, **using a physics-informed data sampling approach**. Performed detailed study of effects with respect to dataset size, data quality, convergence, etc. **Achieved 60-80% error reduction** for tested IC packages compared to conventional surrogate modeling techniques.
- **Developed general-purpose data sampling workflow** for surrogate model training based on observed package sensor data, allowing accurate inference of underlying physics behavior. **Achieved 60-80% error reduction** for tested IC package.
- **Developed Bayesian inference analysis pipeline** to create robust, adaptive data sampling method capable of responding in real-time to live IC sensor data. Invention disclosure/patent in progress.
- Leveraged **PyCharm for code development** to streamline collaboration, testing, and deployment.

**Researcher, Computational Physics/Atomistic Simulation** — ***Toronto, Canada***
***Toronto SickKids Hospital:** SickKids Research Institute* — ***Jan 2015 — Sep 2019***
- **Numerically simulated atomic interactions and protein chemical structures using molecular dynamics** on the SciNet supercomputer cluster. **Analyzed, preprocessed, and visualized large-scale datasets using Python.** Studied small brain peptides and interactions with amyloid-beta oligomers in the context of Alzheimer's disease.
- **Identified and categorized distinct preferential binding modes** consistent with previous literature on amyloid-beta oligomer behavior. Characterized binding propensities using statistical methods.

**Research Assistant, Computational Solid State Physics** — ***Toronto, Canada***
***University of Toronto**: Department of Physics* — ***May 2018— Aug 2018***
- **Constructed new simplified tight-binding model of nickel (II) iodide** by calculated tight-binding parameters for electron-hopping model. Performed numeric simulation and data analysis of electron-hopping models using Python via Monte Carlo simulation and statistical methods.
- **Generated datasets from density functional theory (DFT) using HPC tools** on SciNet supercomputer cluster.

# Fangwei Chang

University of Toronto
fangwei.chang@mail.utoronto.ca

Google Scholar: link to publications
linkedin.com/in/fangwei-c

- **Used Python to calculate energy band structure from new tight-binding model** and confirm agreement with experimental data from literature. Identified distinct magnetic ordering cases, including helimagnetic ordering, via simulated annealing using Monte-Carlo methods.

**Research Assistant, Data Quality and Output** — ***Sudbury, Canada***
***SNOLAB**: SNO+ Experiment* — ***May 2017— Aug 2017***
- **Developed and debugged C++ scripts for data quality checks** on SNO+ detector output ROOT and relevant physics libraries.
- **Assembled and tested calibration apparatus** for SNO+ detector per clean room protocols. Assisted with equipment assembly in ultra-clean underground facility at 2 km depth.

**Research Assistant, Atomic Physics** — ***Toronto, Canada***
***University of Toronto**: Department of Physics* — ***May 2016— Jun 2016***
- **Used RF equipment to test a small solid-state NMR apparatus** to measure precession frequencies of magnetic dipole moments in Pb-207 nuclei, in the context of searching for violations in time reversal symmetry.
- **Designed and built RF filters** and relevant RF setup for experiment.

## Academic Publications

**Wireless Power Transfer Using a Cavity-Enclosed Impedance Metawire Structure**
*2026 IEEE International Symposium on Antennas and Propagation, Detroit, United States*

**An Aperiodic Sub-Wavelength Dipole Structure for Enhancing Near-Field Wireless Power Transfer, *link***
*IEEE Antennas and Wireless Propagation Letters (AWPL), Vol. 23, No. 6, July 2024*

**A Shifted-Beam Method for Near-Field Wireless Power Transfer using Parasitic Arrays, *link***
*2023 IEEE International Symposium on Antennas and Propagation, Portland, United States*

**Shifted-Beam Array Coil for Highly Focal Transcranial Magnetic Stimulation, *link***
*2021 IEEE International Symposium on Antennas and Propagation, Singapore*

## Organizations and Volunteering

**Vice Chair, University of Toronto IEEE Antenna and Propagation Society (AP-S) Student Chapter** — ***Oct 2023 — Dec 2025***
- Hosted social and academic events for U of Toronto IEEE AP-S student chapter. Elected position.

**Team Lead & Composites Fabricator, Rocketry Division, University of Toronto Aerospace Team (UTAT)** — ***Sep 2015 — Aug 2021***
- Led a team of more than 10 members for rocketry airframe subsystem and solid-fuel rocket program (2019-2021).
- Used machine tools and composite manufacturing techniques to design and fabricate by hand carbon fiber airframes for low-altitude sounding rockets. Performed compression testing to verify performance. Coordinated design and construction of solid-fuel rockets for systems testing.

**Social Events Officer, ECE Graduate Student Society (ECEGSS), University of Toronto** — ***Oct 2019 — Sep 2020***
- Organized monthly social and networking events for Electrical and Computer Engineering Graduate Student Society (ECEGSS) at the University of Toronto. Elected position.

**Vice President, University of Toronto Physics Student Union (PhySU)** — ***May 2016 — Apr 2018***
- Managed social and academic events for Physics Student Union. Elected position.

**Research Volunteer, Trauma and Acute Care Surgery, Toronto St. Michael's Hospital** — ***Jun 2015 — Aug 2016***
- Assisted with conducting trauma studies as member of on-call team at St. Michael's Hospital emergency department. Screened and enrolled eligible patients with emergency room staff.

## Citizenship/Legal Status

- Canadian citizen.